Ponavljanje

idealni plin: $pV = RT$ $[J/kg]$

$pV^{n} = konst.$ $0 \leq n \leq \infty$

$\rightarrow n = 0$; $p = konst.$ IZOBARNO $c_n = c_p$

$n = 1$; $pV = konst$ IZOTERMNO $c_n = \infty$

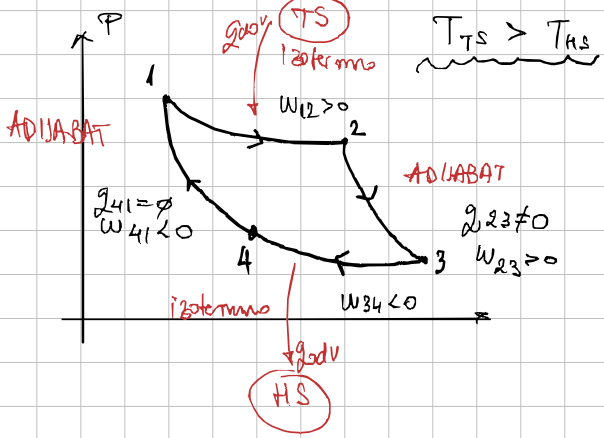
...

$n = \kappa$; $q = \emptyset$ ADIJABATSKA PROMJENA $\kappa = \frac{c_p}{c_v} \rightarrow c_n = 0$

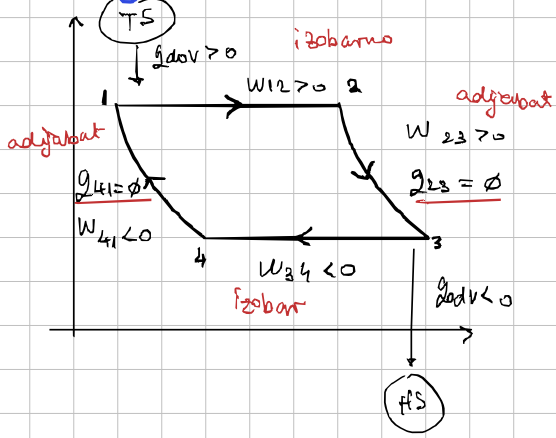
$n = \infty$; $V = konst$ IZOHORNO $c_n = c_v$

$n = n$ POLITROPSKA $c_n = c_v \frac{n - \kappa}{n - 1}$

Carnotov DESNOkretni KP



Brayton-Jouleov DESNO



AV - 02

Zadatak 1.)

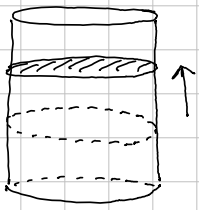
Unutar cilindra sa stapom nalazi se 1m³ idealnog plina na tlaku 800kPa i temperaturi 80C. Plin se izotermno zagrijava do trenutka kada mu se volumen poveća dvostruko. Nakon toga se adijabatski komprimira do početnog volumena.

1) • Kolika se toplina dovede plinu za vrijeme izotermne ekspanzije?

2) • Koliki je tlak nakon adijabatske kompresije?

3) • Koliki se rad mora uložiti za kompresiju plina za vrijeme adijabatske kompresije?

Za idealni plin su poznati podaci Cp = 1004, 87 ~~kJ/kgK~~ J/kgK, Cv = 717.8 ~~kJ/kgK~~ J/kgK.

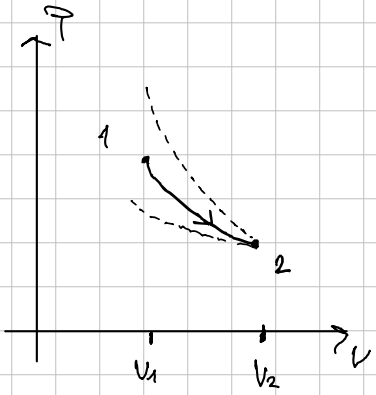


$V_1 = 1\ m^3$

$p_1 = 800\ kPa$

$T_1 = 80°C = 353.15\ K = konst.$

$V_2 = 2V_1$



1) Izotermna ekspanzija $Q_{12} = ?$

$RT \ln \frac{V_2}{V_1} = pV \ln \frac{p_1}{p_2} = Q_{12} \longrightarrow RT \ln \frac{V_2}{V_1} = Q_{12} = \dots \Rightarrow \boxed{Q_{12} = 2035.25\ J}$

$Q_{dov} > 0$

2) $\left(\frac{V_2}{V_1}\right)^{n-1} = \left(\frac{p_1}{p_2}\right)^{\frac{n-1}{n}}$ $\quad * (p \cdot V^n = konst.)$

$n = 1$ za izotermno $\quad C_n = \infty$

$p_1 V_1 = p_2 V_2 \dots > p_2 = 400\ kPa$

$n = K$ za adijabatsku $\quad Q_{12} = 0, \quad n = K = \frac{C_p}{C_v} \Rightarrow C_n = 0$

Adijabatska: $n = K, \ Q_{12} = 0 \quad K = \frac{C_p}{C_v} \rightarrow K = 1.4$

$\left(\frac{V_1}{V_2}\right)^{n-1} = \left(\frac{p_2}{p_3}\right)^{\frac{n-1}{n}} \rightarrow \left(\frac{V_1}{V_2}\right)^{0.4} = \left(\frac{p_2}{p_3}\right)^{\frac{0.4}{1.4}} = \dots \Rightarrow \boxed{p_3 = 1055.6\ kPa}$

3) Uloženi rad $\Rightarrow W_t = ?$ $\quad \cancel{W_{t12} = -C_p \cdot \Delta T}$ tjekom adijabat. komp.

$\frac{T_1}{T_3} = \left(\frac{V_1}{V_2}\right)^{0.4} \longrightarrow T_3 = 465.98\ K$

$\cancel{W_{t12} = -C_p (T_3 - T_2)}$

! $W_{23} = -C_v \cdot \Delta T$

$W_{t12} =$

$R = C_p - C_v = 287\ J/kgK$

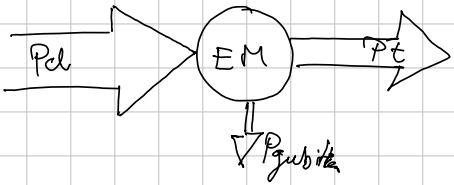
* OVO su formule J/kg znači trebamo masu!

$pV = mRT \rightarrow m = \frac{p_1 \cdot V_1}{RT_1} \Rightarrow [kg] = \frac{[Pa] \cdot [m^3]}{[J/kgK] \cdot [K]} = \frac{[Pa][m^3][kg]}{[J]}$

$m = 7.89\ kg \longrightarrow W_{23} = -m \cdot C_v \cdot \Delta T = -7.89 \cdot 717.8 \cdot (465.98 - 353.15)$

$\boxed{W_{23} = -639.9\ kJ}$

Zadatak 2.)



EM izlazne snage 400 kW (meh snaga na osovini) i stupnja djelovanja ni=0.96 hladi se zrakom koji ventilator tjera kroz kućište. Koliki je maseni protok zraka potreban ako je ulazna temp zraka 25*C, a najviša dozvoljena izlazna temp 50°C? Stupanj djelovanja EM definiran je kao omjer izlazne (meh) snage i uložene električne snage. Specifični je toplinski kapacitet zraka cp= 1005 J/kgk.

$P_{meh} = 400\,kW$ $\quad T_1 = 25°C$ $\quad c_p = 1005\ J/kgK$

$\eta = 0.96$ $\quad T_2 = 323.15$ (max)

$\dot{m} = ?$

$$\eta = \frac{P_{meh}}{P_{el}} = \frac{(korisna)}{(uložena)} \qquad P_{el} = P_{meh} + P_{gub}$$

$$P_{meh} = \eta\,(P_{meh} + P_{gub}) \Rightarrow \dots \quad P_{gub} = P_{meh}\left(\frac{1}{\eta} - 1\right) = 16.67\,kW$$

Zrak se uslijed toplinskih gubitaka zagrijava ?

↳ Zagrijavanjem zraka povećava se entalpija zraka $\Rightarrow P_{gub} = \dot{m}\,\Delta h$

maseni protok zraka

promjena entalpije

$$\Delta h = c_p \cdot \Delta T \longrightarrow P_{gub} = \dot{m} \cdot c_p \cdot (T_2 - T_1)$$

$$[J/s] = [kg/s] \cdot \left[\frac{J}{kg}\right]$$

$$\dot{m} = \frac{P_{gub}}{c_p \cdot (T_2 - T_1)} = \dots$$

kad promatramo otvoreni sustav imamo Δh i on je uvijek $c_p \cdot (T_{kon} - T_{poč})$

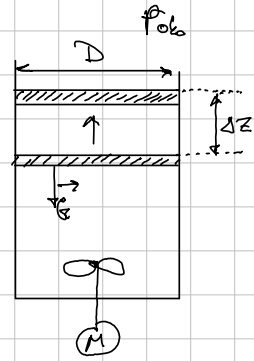
i imamo $\Delta U = c_v\,(T_{kon} - T_{poč})$

$$\boxed{\dot{m} = 0.66\ kg/s}$$

Zadatak 3.)

EM miješa medij (idealni plin) u adijabatskom cilindru promjera 300mm. Motor radi na bateriji 12V i troši 3A. Nakon 50s miješanja pomični poklopac cilindra mase 30kg podigne se za 100mm. Tlak oko cilindra iznosi 95kPa.

Koliko iznpsi promjena unutrašnje kaloričke energije plina?



$D_1 = 300\,mm = 180\,mm$

$\Delta U = G \cdot \Delta T$

$U_m = 12\,V$, $J_m = 3A$ $\Big\}$ $P_m = 36\,\frac{J}{s}$

$Q = \emptyset$ adijabatski proces

→ imamo mehanički rad promjene volumena

↳ rad ovim sustavom = 2 vada se vrše

$t = 50\,s$

$m = 30\,kg$

$\Delta z = 100\,mm$

$\cancel{Q_{12}}^{\,0} = W_{12} + \Delta U$

$p_{okoline} = 95\,kPa$

$\Delta U = ?$

$\Delta U = -W_{12} = -(W_{tr.} + W_{tret} + W_{ok})$

- $W_{tr.}$ → propeler i molekule
- W_{tret} → podižanje poklopca
- W_{ok} → vanjski tlak na poklopac → tlak okolice jednako raspoređen po tijelu

*nema gubitaka

↳ rad koji obavlja EM na račun elektr. energije

→ rad se dovodi izvana pa je < 0

$W_{tr} = -W_{mek} = -W_{el} = -U \cdot I \cdot \Delta t = -1800\,J$

$W_{tret} = G \cdot \Delta z = m \cdot g \cdot \Delta z = \dots = 29.43\,J$

$W_{ok} = p_{ok} \cdot \Delta V = p_{ok} \cdot r^2 \pi \cdot \Delta z = 95 \times 10^3 \cdot 0{,}007 \longrightarrow W_{ok} = 671{,}5\,J$

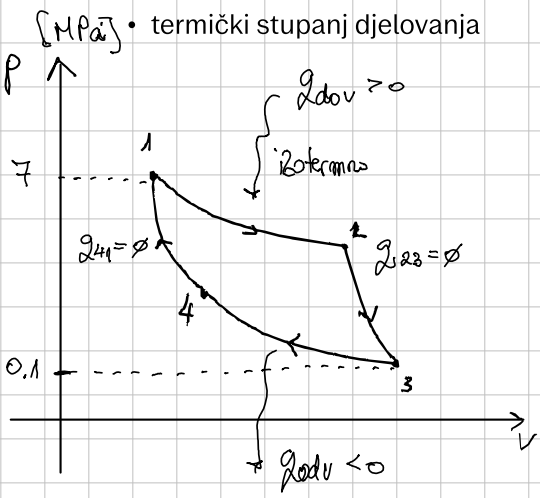
$$\boxed{\Delta U = 1800\,J - 29.43\,J - 671.5\,J = \dots = 1099.1\,J}$$

Unutrašnja kalorička en. raste jer en. koja je dovedena u sustav je zbog trenja veća od one određene zbog rada podižanja tereta i okolice

Zad. 4.)

Temperature toplog i hladnog spremnika Desnokretnog Carnotovog kružnog procesa su 740K i 300K. Tlak plina nakon adijabatske kompresije je 7MPa,a nakon adijabatske ekspanzije 0.1 MPa. Radni fluid je 5kg idealnog plina. Plinska konstanta je 287 J/kgK, a adijabatski indeks k= 1.4. Izračunajte:

- tlakove i volumene plina u 4 karakteristične točke procesa
- dovednu i odvedenu toplinsku energiju te dobiveni mehanički rad
- termički stupanj djelovanja



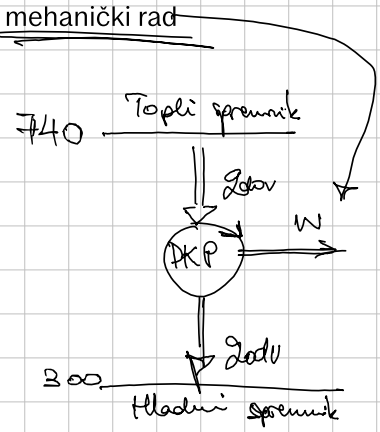
$m = 5kg$

$p_1 = 7MPa$

$p_3 = 0.1 MPa$

$k = 1.4$

$R = 287 \, J/kgK$



$T_{2,4} = ?$ $\quad q_{dov}, q_{odv}, w_t = ?$

$V_{1,2,3,4} = ?$ $\quad \eta = ?$

1) 1-2 IZOTERMNO $n=1$

$p_1 V_1 = p_2 V_2$, $T_1 = T_2 = 740K$

$\frac{V_1}{V_2} = \frac{p_2}{p_1} \Rightarrow \dots \quad \frac{V_1}{V_2} = 0.337$

$q_{dov} = RT \ln \frac{V_2}{V_1} = RT \cdot \ln\left(\frac{175}{59}\right)$

$q_{dov} = 230.9 \, J/kg$ $\times 5kg = 1.155 MJ$

w_{12}

$p_1 V_1 = mRT_1 \rightarrow V_1 = 0.1517 m^3$

2) 2-3 ADIJABTSKI $n = k = \frac{c_p}{c_v} \rightarrow 1.4 c_v = c_v + R$

$0.4 c_v = R \rightarrow c_v = 717.5 \, J/kgK$

$c_p = 1004.5 \, J/kgK$

$\frac{T_{12}}{T_{34}} = \left(\frac{p_2}{p_3}\right)^{\frac{0.4}{1.4}} \rightarrow p_2 = 2.36 MPa$

$V_2 = 0.45 m^3$

~~$W_{t_{23}} = -m \cdot c_p \cdot \Delta T = -m \cdot c_p \cdot (T_3 - T_2) = \dots$~~

$\left(\frac{p_2}{p_3}\right)^{\frac{0.4}{1.4}} = \left(\frac{V_3}{V_2}\right)^{0.4} \rightarrow \left(\frac{p_2}{p_3}\right)^{\frac{1}{1.4}} = \frac{V_3}{V_2} \dots \rightarrow \frac{V_3}{V_2} = 9.56$

3) 3-4 IZOTERMNO $n=1$

w_{34}

$q_{odv} = \dots = -94.59 kJ$ $\times 5kg = -0.473 MJ$

$p_3 V_3 = p_4 V_4$ $\quad T_3 = T_4 = 300K$

$\frac{V_4}{V_3} = \frac{p_3}{p_4} = \dots \Rightarrow \frac{V_4}{V_3} = 0.33\dot{3}$

$p_3 V_3 = mRT_3 \rightarrow V_3 = 4.31 m^3$

$V_4 = 1.43 m^3$

4) 4-1 ADIJABATSKI $n = k = 1.4$

~~$w_t = -m \cdot c_p \cdot \Delta T$~~

$\frac{T_4}{T_1} = \left(\frac{p_4}{p_1}\right)^{\frac{0.4}{1.4}} \rightarrow p_4 = 0.3 MPa$

~~$w_{t_{41}} = -2.21 MJ$~~

ADIABATSKA $\quad W_{23} = \int_{V_2}^{V_3} p\,dV = -m\int_{u_2}^{u_3} du = -m\int_{T_2}^{T_3} c_v\,dT$

$W_{23} = -m \cdot c_v\,(T_3 - T_2) \dots \rightarrow \boxed{W_{23} = 1.579\ MJ}$

isti način za $W_{41} \longrightarrow \boxed{W_{41} = -1.579\ MJ}$

$W_{uk} = W_{12} + W_{23} + W_{34} + W_{41} = 1.156 + \cancel{1.579} - 0.469 - \cancel{1.579} = \boxed{0.687\ MJ}$

$\eta_T = \frac{W}{q_{dov}} = \frac{0.687}{1.156} \rightarrow \boxed{\eta_T = 0.594}$

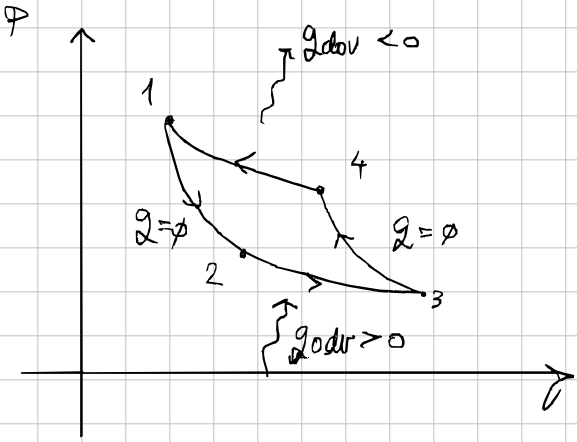
Zadatak 5.)

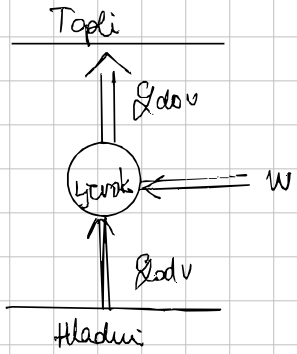
U DKP ⟶ korisna en. = dobiveni meh rad

LJKP ⟶ korisni rad = dovedena toplina ili odvedenoj toplini

Ljevokretnim Carnotovim KP grije se kuća na konstantnu temp 22*C topinom iz rijeke na temp 4*C.

Koliki je iznos dobavljene topline za grijanje i koliki je učinak dizalice topline (faktor preobrazbe) ako utrošak meh rad iznosi 14MJ i uz pretpostavku da se radi o idealnom KP?





$T_{dov} = 295.15\ K$ $\quad W_{meh} = -14MJ$ $\quad q_{dov} = ?$

$T_{odv} = 277.15\ K$ $\quad$ (jer je rad sustavom) $\quad \left|\frac{q_{dov}}{w}\right| = ?$

1-2) ADIJABATSKI $\quad q_{12} = \emptyset$, $n = \kappa$

$$w_{12} = -c_v \cdot (277 - 295) = 18 \cdot c_v$$

$$* \ W_{meh} = \sum W_{sustava}$$

2-3) IZOTERMNO $\quad p_2 V_2 = p_3 V_3 \quad n = 1$

$$\underbrace{w_{23} = q_{23} = RT \ln \frac{V_3}{V_2} = RT \ln \frac{p_2}{p_3}}_{q_{odvedeno}}$$

$$W_{meh} = -\cancel{c_v (T_2 - T_1)} + RT_{dov} \ln \frac{p_2}{p_3} - \cancel{(-c_v)(T_2 - T_1)} + RT_{dov} \ln \frac{p_4}{p_1}$$

$$W_{meh} = RT_{odv} \cdot \ln \frac{p_2}{p_3} + RT_{dov} \cdot \ln \frac{p_4}{p_1}$$

budući da se radi o zatvorenom procesu $\frac{V_2}{V_1} = \frac{V_3}{V_4}$

$$\frac{V_2}{V_3} = \frac{V_1}{V_4} \rightarrow \frac{p_3}{p_2} = \frac{p_4}{p_1}$$

$$W_{meh} = R \cdot \ln \frac{p_2}{p_3} (T_{odv} - T_{dov})$$

$$\Rightarrow \frac{|q_{dov}|}{|W_{meh}|} = \frac{\cancel{R} T_{dov} \cancel{\ln \left(\frac{p_2}{p_3}\right)}}{\cancel{R} \cdot \cancel{\ln \left(\frac{p_2}{p_3}\right)} (T_{odv} - T_{dov})} = \frac{295.15}{|277.15 - 295.15|} \rightarrow \boxed{\frac{|q_{dov}|}{|W_{meh}|} = 16.13}$$

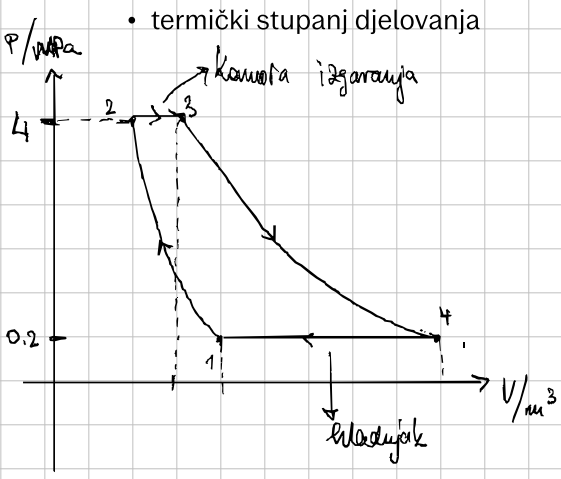
$$|q_{dov}| = 14 \times 10^6 \cdot 16{,}13 \rightarrow |q_{dov}| = 225.82\ MJ \longrightarrow \boxed{q_{dov} = -225.82\ MJ}$$

< 0

Zadatak 6.)

U termoelektrani s pinskom turbinom odvija se idealni Desnokretni Brayton-Jouleov KP. Tlak u komori izgaranja je 4MPa, a u hladnjaku 0.2 MPa. Temp plina na izlazu iz komore izgaranja je 1400K, a na izlazu iz hladnjaka 400K. Pretpostaviti da se proces odvija s idealnim plinom. Plinska konst je 287 J/kgK, a adijabatski indeks k = 1.4. Izračunajte:

- temp i spec volumene plina u 4 karakt točke
- dovedenu i odvedenu toplinsku energiju
- dobiveni meh rad u turbini i uloženi meh rad kompresora
- termički stupanj djelovanja



$p_2 = p_3 = 4\,MPa$ $\quad R = 287\ J/kgK$

$p_4 = p_1 = 0.2\,MPa$ $\quad k = 1.4$

$T_3 = 1400K$

$T_1 = 400K$

1-2 ADIJABATSKA kompresija $q_{12} = \emptyset$, $n = k = 1.4$ → $c_p = 1004.5\ J/kgK$, $c_v = 717.5\ J/kgK$

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{0.4} = \left(\frac{p_1}{p_2}\right)^{\frac{0.4}{1.4}} \rightarrow \boxed{T_2 = 941.42K}$$

specifični volumen

$$v_1 = \frac{RT_1}{p_1}$$

$$\boxed{v_1 = 0.574\ m^3/kg}$$

$$\boxed{v_2 = 0.068\ m^3/kg}$$

$$\frac{V_2}{V_1} = 0.1177$$

$$w_{meh_{12}} = -c_v \cdot \Delta T = \dots \Rightarrow \boxed{w_{meh_{12}} = -388.47\ kJ/kg}$$

$$w_{turbina} = -c_p \Delta T = \dots \Rightarrow \boxed{w_{\cancel{t}} = -543.86\ kJ/kg}$$

komora!

2-3 izobarna ekspanzija $n = \emptyset$, $p = konst.$ $\quad c_n = c_p$ $\quad$ komora izgaranja

$$\boxed{q_{23} = c_p(T_3 - T_2) = 460.64\ kJ}$$

dovedena

$$w_{meh_{23}} = p(V_3 - V_2) = R(T_3 - T_2)$$

$$\boxed{w_{meh_{23}} = 131.61\ kJ/kg}$$

3-4 ADIJABATSKA ekspanzija $q_{34} = \emptyset$

$$\frac{T_3}{T_4} = \left(\frac{p_3}{p_4}\right)^{\frac{0.4}{1.4}} \rightarrow \boxed{T_4 = 594.85\ K}$$

$$\boxed{w_{meh_{34}} = 577.7\ kJ/kg}$$

$$\boxed{w_{t_{34}} = 808.77\ kJ/kg}$$

$$\frac{V_4}{V_3} = 8.5$$

$$v_3 = \frac{RT_3}{p_3} \rightarrow \boxed{v_3 = 0.1\ m^3}$$

$$\boxed{v_4 = 0.854\ m^3}$$

4-1 izobarna kompresija $n = \emptyset$ $\quad p = konst$ $\quad$ HLADNJAK

$$q_{41} = q_{odv} = c_p(T_1 - T_4) \rightarrow \boxed{q_{odv} = -195.73\ kJ/kg}$$

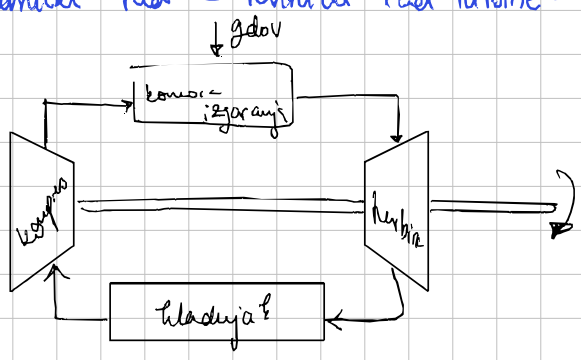
$$w_{meh_{41}} = R(T_1 - T_4) \rightarrow \boxed{w_{meh_{41}} = -55.91\ kJ/kg}$$

$$w_{meh} = \sum w_{meh_i} = -388.47 + 131.61 + 577.7 - 55.91$$

$$\boxed{w_{meh} = 264.93\ kJ/kg}$$

$$\Delta w_{tehnički} = w_{t_{12}} + w_{t_{34}} \rightarrow \boxed{\Delta w_{tehnički}\ 264.91\ kJ/kg}$$

Mehanički rad = tehnički rad turbine − tehnički rad kompresora



$$\eta_t = \frac{w_{meh}}{q_{dovedena}} = \frac{264.9}{460.64\ kJ}$$

$$\boxed{\eta_t = 0.575}$$